Name

Mathematics Advanced Year 12

Differentiating Logarithmic Functions

Book 2

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Contents

Syllabus Content 2
Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-12-04 selects and applies differentiation methods to solve problems

Differentiation with logarithmic functions

- Use chain rule on the identity $e^{\ln x} = x$ to prove the formula $\frac{d}{dx}(\ln x) = \frac{1}{x}$ for the derivative of the natural logarithm function where $x > 0$
- Prove and use the formula $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$, where a is a constant and $a > 0$
- Use the chain rule to differentiate functions of the form $\ln f(x)$

Derivative of Natural Logarithmic Functions

Investigation Derivative of natural logarithmic functions.

In this investigation we will derive the derivative of $y = \ln x$

We will start with $y = e^{\ln x}$, differentiate by letting $u = \ln x$:

$$y = e^{\ln x}$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Let $u = \ln x$

$$\frac{du}{dx} = u'(x)$$

(where $u'(x)$ is the derivative of $\ln x$, which we want to find)

Let $y = e^u$

$$\frac{dy}{du} =$$

Now, we apply the identity $e^{\ln x} = x$.

From above, we can rewrite:

$$\frac{dy}{dx} = u'(x)e^{\ln x}$$

$$=$$

And at the same time, using $y = e^{\ln x} = x$,

$$\frac{dy}{dx} =$$

Equating the two expressions we have for $\frac{dy}{dx}$, we can find an expression for $u'(x)$, which is the derivative of $\ln x$.

ⓘ Derivative of Natural Logarithmic Function

$$f(x) = \ln x$$

$$f'(x) = \frac{1}{x}$$

Identify the derivative of a logarithmic function.

$f(x) = 2\ln x$ $f'(x) = 2 \times \frac{1}{x}$ $= \frac{2}{x}$	$f(x) = x^2 + \ln x$ $f'(x) = 2x + \frac{1}{x}$	$f(x) = 5x^2 - 3\ln x$ $f'(x) = 10x - \frac{3}{x}$

Differentiate these functions.

a $f(x) = 5\ln x$	**b** $f(x) = -5\ln x$	**c** $f(x) = x - 5\ln x$
d $f(x) = 3 - 5\ln x$	**e** $f(x) = x^3 - 5\ln x$	**f** $f(x) = 7x^2 + 12\ln x$

ⓘ Investigation Chain rule for logarithmic functions.

Consider $y = \ln(f(x))$.

Using chain rule, find the general form of the derivative.

$y = \ln(f(x))$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

Let $u = f(x)$

$\frac{du}{dx} =$

Let $y = \ln u$

$\frac{dy}{du} =$

ⓘ Chain Rule for Natural Logarithmic Function

$$y = \ln f(x)$$
$$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$$

Identify the derivative of a logarithmic function.

$f(x) = \ln(7x)$ $f'(x) = \frac{7}{7x}$ $= \frac{1}{x}$	$f(x) = \ln(x^2)$ $f'(x) = \frac{2x}{x^2}$ $= \frac{2}{x}$	$f(x) = \ln(3x^5 + 7x)$ $f'(x) = \frac{15x^4 + 7}{3x^5 + 7x}$

Explain why the derivative of $\ln(7x)$ is the same as the derivative of $\ln(x)$.

Explain why the derivative of $\ln(x^2)$ is the same as the derivative of $2\ln(x)$.

Differentiate these functions.

a $f(x) = \ln 7x$	**b** $f(x) = \ln 7 + \ln x$	**c** $f(x) = \ln(5x^2)$
d $f(x) = \ln\left(\frac{5}{x}\right)$	**e** $f(x) = \ln(\sqrt{x})$	**f** $f(x) = \ln(11x^3 - 7x^2)$

ⓘ The Log Laws

Inverse Relationship	$\log_a a^x = a^{\log_a x} = x$
Log Law of Multiplication	$\log_a(xy) = \log_a x + \log_a y$
Log Law of Division	$\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$
Log Law of Indices	$\log_a(x^n) = n\log_a x$

Investigation using log laws to simplify before you differentiate.

We can use log laws to change complex logarithms to multiples or sums of simpler logs.

Simplify, then differentiate these functions.

a $f(x) = \ln(7x^2)$

b $f(x) = \ln((3x-7)^5)$

c $f(x) = \ln(x^3)$

d $f(x) = (\ln x)^3$

e $f(x) = \ln\left(\frac{1+x}{1-x}\right)$

f $f(x) = \frac{\ln(1+x)}{1-x}$

Example Differentiate natural logarithmic functions.

Differentiate

$y = x^3 \ln x$ use product rule

$y' = u'v + uv'$

$= 3x^2 \ln x + x^3 \cdot \frac{1}{x}$

$= x^2(3 \ln x + 1)$

Differentiate

$y = \frac{\ln(1+x)}{x}$ use quotient rule

$y' = \frac{u'v - uv'}{v^2}$

$= \frac{\frac{1}{1+x}(x) - \ln(1+x)\,(1)}{x^2}$

$= \frac{\frac{x}{1+x} - \ln(1+x)}{x^2}$

$= \frac{x - (1+x)\ln(1+x)}{x^2(1+x)}$

Practice

Differentiate

$y = x^4 \ln x$

$x^3(4 \ln x + 1)$

Differentiate

$y = \frac{\ln(2x)}{x^3}$

$\frac{1 - 3\ln(2x)}{x^4}$

Differentiate

$y = (\ln(x) + x)^5$

$5(x + \ln(x))^4\left(\frac{1+x}{x}\right)$

Differentiate

$y = \frac{3x}{\ln x}$

$\frac{3(\ln x - 1)}{(\ln x)^2}$

1 Differentiate:

a $y = \log_e(x + 2)$

$y' = \frac{1}{x + 2}$

b $y = \log_e(x - 3)$

$y' = \frac{1}{x - 3}$

c $y = \log_e(3x + 4)$

$y' = \frac{3}{3x + 4}$

d $y = \log_e(2x - 1)$

$y' = \frac{2}{2x - 1}$

e $y = \log_e(-4x + 1)$

$y' = -\frac{4}{-4x + 1}$

f $y = \log_e(-3x + 4)$

$y' = -\frac{3}{-3x + 4}$

g $y = \ln(-2x - 7)$

$y' = -\frac{2}{-2x - 7} = \frac{2}{2x + 7}$

h $y = 3\ln(2x + 4)$

$y' = 3\left(\frac{2}{2x + 4}\right) = \frac{6}{2x + 4} = \frac{3}{x + 2}$

i $y = 5\ln(3x - 2)$

$y' = 5\left(\frac{3}{3x - 2}\right) = \frac{15}{3x - 2}$

j $y = \ln 2x$

$y' = \frac{1}{x}$

k $y = \ln 5x$

$y' = \frac{1}{x}$

l $y = \ln 7x$

$y' = \frac{1}{x}$

m $y = 4\ln 7x$

$y' = \frac{4}{x}$

n $y = 3\ln 5x$

$y' = \frac{3}{x}$

o $y = 4\ln 6x$

$y' = \frac{4}{x}$

2 Find $\frac{dy}{dx}$ for each function. Then evaluate $\frac{dy}{dx}$ at $x = 3$.

a $y = \log_e(2x - 1)$

$y' = \frac{2}{2x - 1}$,
$y'(3) = \frac{2}{5}$

b $y = \log_e(4x + 3)$

$y' = \frac{4}{4x + 3}$,
$y'(3) = \frac{4}{15}$

c $y = 6\ln(2x + 9)$

$y' = \frac{12}{2x + 9}$,
$y'(3) = \frac{4}{5}$

3 Differentiate these functions.

a $2 + \log_e x$

$\frac{1}{x}$

b $5 - \log_e(x + 1)$

$-\frac{1}{x + 1}$

c $x + 4\log_e x$

$1 + \frac{4}{x}$

d $2x^4 + 1 + 3\log_e x$

$8x^3 + \frac{3}{x}$

e $\ln(2x - 1) + 3x^2$

$\frac{2}{2x - 1} + 6x$

f $x^3 - 3x + 4 + \ln(5x - 7)$

$3x^2 - 3 + \frac{5}{5x - 7}$

4 Use the log laws to simplify each function, then differentiate it.

a $y = \ln x^3$

$y = 3\ln x$, $y' = \frac{3}{x}$

b $y = \ln x^2$

$y = 2\ln x$, $y' = \frac{2}{x}$

c $y = \ln x^{-3}$

$y = -3\ln x$, $y' = -\frac{3}{x}$

d $y = \ln x^{-2}$

$y = -2\ln x$, $y' = -\frac{2}{x}$

e $y = \ln\sqrt{x}$

$y = \frac{1}{2}\ln x$, $y' = \frac{1}{2x}$

f $y = \ln\sqrt{x} + 1$

$y = \frac{1}{2}\ln(x + 1)$, $y' = \frac{1}{2(x + 1)}$

5 Differentiate these functions.

a $y = \log_e \frac{x}{2}$

$\frac{1}{x}$

b $y = \log_e \frac{x}{3}$

$\frac{1}{x}$

c $y = 3\log_e \frac{x}{5}$

$\frac{3}{x}$

d $y = -6\log_e \frac{x}{5}$

$-\frac{6}{x}$

e $y = x + \log_e\left(\frac{1}{3}x\right)$

$1 + \frac{1}{x}$

f $y = 4x^3 - \log_e\left(\frac{1}{5}x\right)$

$12x^2 - \frac{1}{x}$

6 Use chain rule to differentiate:

a $\ln(x^2 + 1)$

$\frac{dy}{dx} = \frac{2x}{x^2 + 1}$

b $\ln(2 - x^2)$

$\frac{dy}{dx} = \frac{-2x}{2 - x^2}$

c $\ln(1 + e^x)$

$\frac{dy}{dx} = \frac{e^x}{1 + e^x}$

d $\log_e(x^2 + 3x + 2)$

$\frac{2x + 3}{x^2 + 3x + 2}$

e $\log_e(1 + 2x^3)$

$\frac{6x^2}{1 + 2x^3}$

f $\ln(e^x - 2)$

$\frac{e^x}{e^x - 2}$

g $x + 3 - \ln(x^2 + x)$

$1 - \frac{2x + 1}{x^2 + x}$

h $x^2 + \ln(x^3 - x)$

$2x + \frac{3x^2 - 1}{x^3 - x}$

i $4x^3 - \ln(2x^2 - 3x + 1)$

$12x^2 - \frac{4x - 3}{2x^2 - 3x + 1}$

7 Find the gradient, and the angle of inclination correct to the nearest minute, of the tangent to $y = \ln x$ at the points where:

a $x = 1$

$y' = \frac{1}{x}$,
at $x = 1$:
$m = 1$,
$\theta = 45°$

b $x = 3$

at $x = 3$:
$m = \frac{1}{3}$
$\theta = 18°26'$

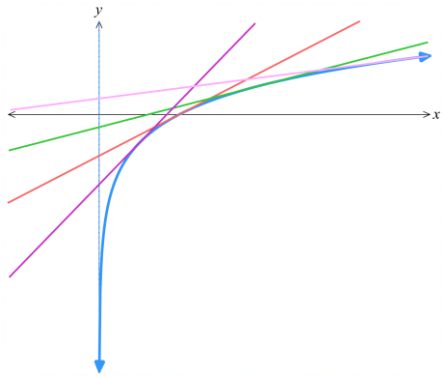
c $x = \frac{1}{2}$

at $x = \frac{1}{2}$
$m = 2$,
$\theta = 63°26'$

d $x = 4$

at $x = 4$:
$m = \frac{1}{4}$
$\theta = 14°2'$

e Draw a sketch of the curve and the four tangents, showing the angles of inclination.



8 Differentiate these functions using the product rule.

a $x \log_e x$

$y' = log_e x + x\left(\frac{1}{x}\right)$
$= 1 + log_e x$

b $x \log_e(2x + 1)$

$y' = log_e(2x + 1) + x\left(\frac{2}{2x + 1}\right)$
$= \frac{2x}{2x + 1} + log_e(2x + 1)$

c $(2x + 1)\log_e x$

$y' = 2\, log_e x + (2x + 1)\left(\frac{1}{x}\right)$
$= \frac{2x + 1}{x} + 2\, log_e x$

d $x^4 \log_e x$

$y' = 4x^3 \log_e x + x^4\left(\frac{1}{x}\right)$
$= x^3(1 + 4\, log_e x)$

e $(x + 3)\log_e(x + 3)$

$y' = \log_e(x + 3) + (x + 3)\left(\frac{1}{x + 3}\right)$
$= log_e(x + 3) + 1$

f $(x - 1)\log_e(2x + 7)$

$y' = log_e(2x + 7) + (x - 1)\left(\frac{2}{2x + 7}\right)$
$= \frac{2(x - 1)}{2x + 7} + log_e(2x + 7)$

g $e^x \log_e x$

$y' = e^x log_e x + e^x\left(\frac{1}{x}\right)$
$= e^x\left(\frac{1}{x} + log_e x\right)$

h $e^{-x} \log_e x$

$y' = -e^{-x} log_e x + e^{-x}\left(\frac{1}{x}\right)$
$= e^{-x}\left(\frac{1}{x} - log_e x\right)$

9 Differentiate these functions using the quotient rule.

a $y = \frac{\log_e x}{x}$

$y' = \frac{x\left(\frac{1}{x}\right) - \log_e x(1)}{x^2}$

$= \frac{1 - \log_e x}{x^2}$

b $y = \frac{\log_e x}{x^2}$

$y' = \frac{x^2\left(\frac{1}{x}\right) - \log_e x(2x)}{x^4}$

$= \frac{1 - 2\log_e x}{x^3}$

c $y = \frac{x}{\log_e x}$

$y' = \frac{\log_e x(1) - x\left(\frac{1}{x}\right)}{(\log_e x)^2}$

$= \frac{\log_e x - 1}{(\log_e x)^2}$

d $y = \frac{x^2}{\log_e x}$

$y' = \frac{\log_e x(2x) - x^2\left(\frac{1}{x}\right)}{(\log_e x)^2}$

$= \frac{x(2\log_e x - 1)}{(\log_e x)^2}$

e $y = \frac{\log_e x}{e^x}$

$y' = \frac{e^x\left(\frac{1}{x}\right) - \log_e x(e^x)}{e^{2x}}$

$= \frac{1 - x\log_e x}{xe^x}$

f $y = \frac{e^x}{\log_e x}$

$y' = \frac{\log_e x(e^x) - e^x\left(\frac{1}{x}\right)}{(\log_e x)^2}$

$= \frac{e^x(x\log_e x - 1)}{x(\log_e x)^2}$

10 Use the log laws to simplify each function, then differentiate it.

a $y = \log_e 5x^3$

$y = \log_e 5 + 3\log_e x,$

$y' = \frac{3}{x}$

b $y = \log_e \sqrt[3]{x}$

$y = \left(\frac{1}{3}\right)\log_e x,$

$y' = \frac{1}{3x}$

c $y = \log_e\left(\frac{3}{x}\right)$

$y = \log_e 3 - \log_e x,$

$y' = -\frac{1}{x}$

d $y = \ln\sqrt{2 - x}$

$y = \frac{1}{2}\ln(2 - x),$

$y' = -\frac{1}{2(2 - x)} = \frac{1}{2x - 4}$

e $y = \log_e\left(\frac{3}{x}\right)$

$y = \log_e 3 - \log_e x$

$y' = \frac{1}{1 + x} + \frac{1}{1 - x}$

f $y = \ln\left(\frac{1 + x}{1 - x}\right)$

$y = \ln(1 + x) - \ln(1 - x),$

$y' = \frac{1}{1 + x} + \frac{1}{1 - x}$

g $y = \log_e 2^x$

$y = x\log_e 2,$

$y' = \log_e 2$

h $y = \log_e e^x$

$y = x,$

$y' = 1$

i $y = \log_e x^x$

$y = x\log_e x,$

$y' = 1 + \log_e x$

8

a Find the tangent to $y = \ln x$ at $P(e, 1)$, and prove that it passes through O.

$y = \ln x, \frac{dy}{dx} = \frac{1}{x}$

At $P(e, 1)$: gradient $= \frac{1}{e}$

Tangent: $y - 1 = \left(\frac{1}{e}\right)(x - e)$

$y = \left(\frac{1}{e}\right)x$

Passes through $O(0, 0)$ as equation is of the form $y = kx$

b Find the tangent to $y = \ln x$ at $Q(1, 0)$, and prove that it passes through $A(0, -1)$.

At $Q(1, 0)$: gradient $= \frac{1}{1} = 1$

Tangent: $y - 0 = 1(x - 1)$

$y = x - 1$

Equation is true when $x = 0, y = -1$, passes through A(0, -1)

c Find the tangent to $y = \ln x$ at $R\left(\frac{1}{e}, -1\right)$, and prove that it passes through $B(0, -2)$.

$At\ R\left(\frac{1}{e}, -1\right)$: gradient $= \frac{1}{\frac{1}{e}} = e$

Tangent: $y - (-1) = e\left(x - \frac{1}{e}\right)$

$y + 1 = ex - 1$

$y = ex - 2$

When $x = 0, y = -2$, passes through $B(0, -2)$

d Find the normal to $y = \ln x$ at $A(1, 0)$, and its y-intercept.

At A$(1, 0)$: gradient of tangent $= 1$

Normal has gradient $= -1$

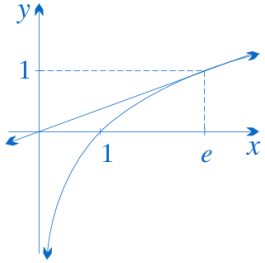
Normal: $y - 0 = -1(x - 1)$

$y = -x + 1$

y intercept is 1

9

a In the previous question you showed that the tangent at $P(e, 1)$ on the curve $y = \ln x$ passes through the origin. Sketch the graph, showing the tangent, and explain graphically why no other tangent passes through the origin.



As P moves left along the curve, the tangent becomes steeper, so it does not pass through the origin.
As P moves right, the angle of the tangent becomes less steep, hence it does not pass through the origin.

b Arguing from the graph, classify the points in the plane according to whether 0, 1 or 2 tangents pass through them.

0 tangents: all points below the curve

1 tangent: points on the curve, and points on or above the curve to the left of the y-axis

2 tangents: points above the curve and right of the y-axis

10 Find, giving answers in the form $y = mx + b$, the equations of the tangent and normal to:

a $y = 4\ln x$ at the point $Q(1, 0)$

$\frac{dy}{dx} = \frac{4}{x}$

$At\ Q(1,0)\text{: gradient} = 4$

$\text{Tangent: } y = 4x - 4$

$\text{Normal: } y = -\frac{x}{4} + \frac{1}{4}$

b $y = \ln x + 3$ at the point $R(1, 3)$

$\frac{dy}{dx} = \frac{1}{x}$

$At\ R(1,3)\text{: gradient} = 1$

$\text{Tangent: } y = x + 2$

$\text{Normal: } y = -x + 4$

c $y = 2\ln x - 2$ at the point $S(1, -2)$

$\frac{dy}{dx} = \frac{2}{x}$

$At\ S(1,-2)\text{: gradient} = 2$

$\text{Tangent: } y = 2x - 4$

$\text{Normal: } y = -\frac{1}{2}x - \frac{1}{2}$

11

a Show that the point $P(1, 0)$ lies on the curve $y = \ln(3x - 2)$.

$\text{When } x = 1\text{: } y = \ln(3(1) - 2) = \ln 1 = 0$

$P(1,0)$ lies on curve

b Find the equations of the tangent and normal at P, and their y-intercepts.

$\frac{dy}{dx} = \frac{3}{3x - 2}$

$At\ P(1,0)\text{: gradient} = \frac{3}{1} = 3$

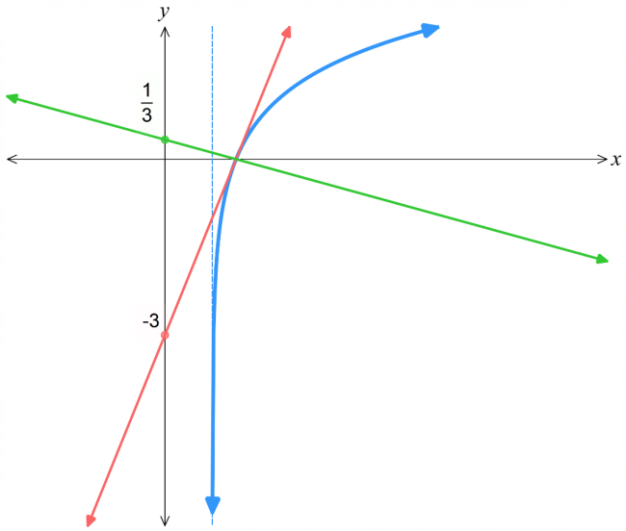
$\text{Tangent: } y = 3x - 3$

$y\text{ intercept: } -3$

$\text{Normal: } y = -\frac{1}{3}x + \frac{1}{3}$

$y\text{ intercept: } \frac{1}{3}$

c Sketch the situation and find the area of the triangle formed by the tangent, the normal and the y-axis.



$\text{Area} = \frac{bh}{2} = \frac{\left(\frac{1}{3} + 3\right)(1)}{2} = \frac{5}{3}\text{ square units}$

11 Differentiate each function using the chain, product or quotient rules.
Then find any values of x for which the derivative is zero.

a $y = x \ln x - x$

$y' = \ln x + x\left(\frac{1}{x}\right) - 1 = ln\, x$

$\ln x = 0,$

$x = 1$

b $y = x^2 \ln x$

$y' = 2x \ln x + x^2\left(\frac{1}{x}\right) = x(1 + 2ln\, x)$

$1 + 2 \ln x = 0,$

$x = e^{-\frac{1}{2}}$

c $y = \frac{\ln x}{x}$

$y' = \frac{1 - ln\, x}{x^2}$

$1 - \ln x = 0,$

$x = e$

d $y = (\ln x)^4$

$y' = 4(ln\, x)^3\left(\frac{1}{x}\right) = \frac{4(ln\, x)^3}{x}$

$\ln x = 0,$

$x = 1$

e $y = (2 \ln x - 3)^4$

$y' = 4(2ln\, x - 3)^3\left(\frac{2}{x}\right) = \frac{8(2ln\, x - 3)^3}{x}$

$2 \ln x - 3 = 0,$

$x = e^{\frac{3}{2}}$

f $y = \frac{1}{\ln x}$

$y' = -\frac{1}{x(ln\, x)^2}$

Never zero

g $y = \ln(\ln x)$

$y' = \frac{1}{x\, ln\, x}$

Never zero

h $y = x \ln x$

$y' = 1 + ln\, x$

$\ln x = -1,$

$x = \frac{1}{e}$

12 Find the coordinates of the point on $y = \ln x$ where the tangent has gradient $\frac{1}{2}$.
Then find the equation of the tangent and normal there, in the form $y = mx + b$.

$\frac{dy}{dx} = \frac{1}{x} = \frac{1}{2}$

$x = 2, y = \ln 2$

Point: $(2, \ln 2)$

Tangent: $y - ln\, 2 = \left(\frac{1}{2}\right)(x - 2)$

$y = \frac{1}{2}x - 1 + \ln 2$

Normal: $y - \ln 2 = -2(x - 2)$

$y = -2x + 4 + \ln 2$

Derivative of Logarithmic Functions of Any Base

Investigation Derivative of logarithmic functions of any base.

We can change the base of any logarithm to base e using the change of base law:

$$\log_a x = \frac{\ln x}{\ln a}$$

We can then differentiate, as $\ln a$ is a constant.

Differentiate:

$f(x) = \log_{10} x$

$f'(x) =$

$f(x) = \log_2 x$

$f'(x) =$

$f(x) = \log_{17} x$

$f'(x) =$

ⓘ Derivative of Logarithms of Any Base

$$y = \log_a x$$

$$\frac{dy}{dx} = \frac{1}{x(\ln a)}$$

Identify the derivative of a logarithmic function.

$f(x) = \log_5 x$ $f'(x) = \frac{1}{x \ln 5}$	$f(x) = \log_8 x$ $f'(x) = \frac{1}{x \ln 8}$	$f(x) = \log_{52} x$ $f'(x) = \frac{1}{x \ln 52}$

Differentiate these functions.

a $f(x) = \log_{11} x$	**b** $f(x) = \log_{62} x$	**c** $f(x) = \log_{\frac{3}{4}} x$
d $f(x) = \log_e x$	**e** $f(x) = 7\log_2 x$	**f** $f(x) = 11\log_5 x$

ⓘ Investigation Chain rule for logarithmic functions of any base.

Consider $y = \log_a(f(x))$

Using chain rule, find the general form of the derivative.

$y = \log_a(f(x))$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Let $u = f(x)$

$$\frac{du}{dx} =$$

Let $y = \log_a u$

$$\frac{dy}{du} =$$

ⓘ Chain Rule for Logarithms of Any Base

$$y = \log_a f(x)$$

$$\frac{dy}{dx} = \frac{f'(x)}{(\ln a) f(x)}$$

Identify the derivative of a logarithmic function.

$f(x) = \log_5(2x)$ $f'(x) = \frac{2}{(\ln 5)(2x)}$	$f(x) = \log_5(2x + 7)$ $f'(x) = \frac{2}{(\ln 5)(2x + 7)}$	$f(x) = \log_7(x^3 - 7x)$ $f'(x) = \frac{3x^2 - 7}{(\ln 7)(x^3 - 7x)}$

Differentiate these functions.

a $f(x) = \log_3(7x)$	**b** $f(x) = \log_3(7x - 3)$	**c** $f(x) = \log_{11}(x^4)$
d $f(x) = \log_{12}(3x^2 + 5)$	**e** $f(x) = \log_7(\sqrt{x})$	**f** $f(x) = \log_{12}\left(\frac{1}{x}\right)$

1 Use the change-of-base formula to express these with base e, then differentiate them.

a $y = \log_2 x$

$$y = \log_2 x = \frac{\ln x}{\ln 2}$$
$$y' = \frac{1}{x \ln 2}$$

b $y = \log_{10} x$

$$y = \log_{10} x = \frac{\ln x}{\ln 10}$$
$$y' = \frac{1}{x \ln 10}$$

c $y = 3\log_5 x$

$$y = 3\log_5 x = \frac{3\ln x}{\ln 5}$$
$$y' = \frac{3}{x \ln 5}$$

2 Differentiate using the formula:

a $y = \log_3 x$

$$y' = \frac{1}{x \ln 3}$$

b $y = \log_7 x$

$$y' = \frac{1}{x \ln 7}$$

c $y = 5\log_6 x$

$$y' = \frac{5}{x \ln 6}$$

3

a Differentiate $y = \log_2 x$. Hence find the gradient of the tangent to the curve at $x = 1$.

$$y = \log_2 x, y' = \frac{1}{x\ ln\ 2}$$
$$\text{At } x = 1\text{: gradient} = \frac{1}{\ln 2}$$

b Hence find the equation of the tangent there.

$$\text{At } x = 1\text{: } y = log_2 1 = 0$$
$$\text{Tangent: } y - 0 = \left(\frac{1}{\ln 2}\right)(x - 1)$$
$$y = \frac{(x - 1)}{\ln 2}$$

4 Find $f'(1)$ if $f(x) = \ln\sqrt{2 - x}$.

$$f'(x) = \left(\frac{1}{2}\right)\left(-\frac{1}{2 - x}\right) = -\frac{1}{2(2 - x)}$$
$$f'(1) = -\frac{1}{2(1)} = -\frac{1}{2}$$

5

a Find the equation of the tangent to $y = \log_{10} x$ at $(10,1)$

$$y' = \frac{1}{x \ln 10}$$
$$\text{At } x = 10\text{: gradient} = \frac{1}{10 \ln 10}$$
$$At\ x = 10\text{: } y = 1$$
$$\text{Tangent: } y - 1 = \left(\frac{1}{10 \ln 10}\right)(x - 10)$$
$$x - 10y \ln 10 - 10 + 10 \ln 10 = 0$$
$$x - 10y \ln 10 + 10(\ln 10 - 1) = 0$$

b At what value of x will the tangent have gradient 1?

$$\frac{1}{x\ ln\ 10} = 1$$
$$x = \frac{1}{\ln 10}$$